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YOUVA

Assignment 5

Q1. Count Digits in a Number

```
psvm(){

  Scanner sc = new Scanner(System.in);
  int n = sc.nextInt(); // 54321
  int count = 0;

   while (n != 0){
      count ++
      n = n/10;
   }
}
```

* dry run

steps	n	count	n/10
1	54321	1	54321 /10 = 5432
2	5432	2	5432/10 = 543
3	543	3	543/10 = 54
2	54	4	54/10 = 5
1	5	5	5/10 = 0

Q2. Extract All Digit (right to left).

```
psvm(){

  int n = 5382;
```

```
while(n!=0){
    int digit = n%10;
    S.O.P(digit);
    n=n/10;
  }
}
```

dry run

Steps	n	prints(n%10)	n/10
1	5382	2	5382/10 = 538
2	538	8	538/10 = 53
3	53	3	53/10 = 5
~~8~~ 4	5	5	5/10 = 0

Q3. Reverse a number.

```
psvm(){

int n = 12345
int rev = 0;

  while (n!=0){
     int digit = n%10;
     rev = rev×10 + digit
     n = n/10
  }
  S.O.P(rev);
}
```

dry run

Steps n

n = 12345 → digit % 10 (4) → rev×10 (0×10) + digit (4) = 4

→ n/10 = 1234 → digit % 10 (3) → rev×10 (4×10=40) + digit (3) = 43

→ n/10 = 12 → digit % 10 (2) → rev×10 (43×10=430) + digit (2) = 432

→ n/10 = 1 → digit % 10 (2) → rev×10 (432×10=4320) + digit (1) = 4321

→ n/10 = 0;

Q4. Palindrome Number.

```
psvm(){
   int n = 121
   int rev = 0;
   int original = n;
   while(n!=0){
      int digit = n%10;
      rev = rev×10 + digit
      n=n%10;
   }
   if(original == rev){
        S.O.P(" It's a palindrome number");
   }else{
        S.O.P("Not a palindrome number");
   }
}
```

Q5 Sum of Digits

```
psum(){

   int n= 543
   int sum=0;

   while(n!=0){
       int digit= n%10;
       sum+= digit;
       n=n/10;
   } SOP(sum);
}
```

dry run	Steps	n	n%10 ~~Sum~~	Sum ~~n%10~~	n/10
	1	543	3	3	543/10=54
	2	54	4	7	54/10=5
	3	5	5	12	5/10=0

Q6. Armstrong number (cube of digit addition= input no.)

```
psum(){
    int n= 153
    int sum=0, original=n;
    while(n!=0){
        int digit= n%10;
        sum= sum+(digit*digit*digit);
        n=n/10;
    }
        if(sum==original){
            S.O.P("Armstrong number");
        }else{
            S.O.P("Not An Armstrong Number");
        }
}
```

#dry run	Steps	n	n%10 ~~Sum~~	Sum ~~n%10~~	n/10
	1	153	3	sum+ 3^3= 27	153/10=15
	2	15	5	sum+ 5^3=125=152	15/10=1
	3	1	1	sum+ 1^3= 1=153	1/10=0

Q7. Print All Divisors

```
psum(){
    int n=12
    while(n
    for(int i=1; i<=math.sqrt(n); i++){
            if(n%i==0){
                S.O.P(i);
            }
    }
}
```

1, 2, 3, 4, 6, 12

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Q8. Count Number of Divisor.

```
psvm(){

    int n = 12
    int count = 0;

    for (int i = 1; i <= math.squrt(n); i++){

            if(n % i == 0){
                count++;
            }
    }
}   S.O.P(" Sum);
}
```

// Output :- 6

Q9 Check Prime Number

```
psvm(){

    int n = 12
    int count = 0;

    for(int i = 0, i <= math.squrt(n), i++){

            if(n % i == 0){
                count++;
            }
    }
```

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```
    if(count == 2){
            S.O.P(" It's a prime")
    }else{
            S.O.P(" It's not a prime")
    }
}
```

o/p : prime

Q10 Print all Prime Numbers up to N

```
psvm (){

    int n = 20;

    for (int i = 1; i <= n; i++){
        int count = 0;

        for (int j = 1; j * j <= i; j++){
            if(i % j == 0){
                count++
            }
        }

        if (count == 2){
                S.O.P(i + " ");
        }
    }
}
```

o/p : 2, 3, 5, 7, 11, 13, 17, 19

Q12 Find GCD / HCF of Two Numbers (Brute Force), using Euclidean Algorithm

```
psvm()

    int a=12
    int b=18
    int gcd=1
        for(int i=1; i<=Math.min(a,b); i++){
            if(a%i==0 && b%i==0){
                gcd=i;
            }
        }
        S.O.P(gcd);
}
```

Output: 6

Q11 Find GCD using brute force.

```
psvm(){

    int a=12;
    int b=18;
    int gcd=1;

    for(int i=1; i<=a && i<=b; i++){
        if(a%i==0 && b%i==0){
            gcd=i;
        }
    }
```

```
        System.out.println("a:"+a+"b:"+b);
    }
```

Q13 Check Coprime Numbers

```
int a=8;
int b=15;

for(int i=1; i<=Math.min(a,b); i++){
        if(a%i==0 && b%i==0){
            gcd=i;
        }
}

    if(gcd==1){
        S.O.P("Coprime");
    }
}
```

Q14. Perfect number.

```
psvm(){
    int n=6

    for(int i=1; i<n; i++){
        if(n%i==0){
            sum=sum+i;
        }
    }
```

```
if(sum==n){
        S.O.P("Perfect number");
}else{
        S.O.P("Not Perfect Number");
}
}
```